



The Undrained Shear Behavior of Reinforced Clayey Sand

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Abstract Undrained instability in sandy soil containing fine particles such as liquefaction has remained a troublesome challenge in geotechnical research. Soil reinforcement could be taken as an approach to enhance its engineering properties like shear strength, cohesion, and bearing capacity. The present paper aims to evaluate the capability of geotextile reinforcement in improving undrained shear strength and diminishing undrained brittleness of loose clayey sands. A series of consolidated undrained triaxial tests were carried out on clayey sands containing 0, 5, 10, 15, 20 and 30% fines content (F.C.) under 100, 150 and 200 kPa confining pressures. To investigate the effect of geotextile reinforcement, experiments were performed in three reinforcement arrangements. The results indicate that geotextile reinforcement increases the steady state shear strength and decreases liquefaction potential and softening behavior. Geotextile reinforcement leads to decrease brittleness index of clayey sand specimens. It can also be pointed out that the presence of clay fines changes the dilative behavior of clean sand to contraction, and consequently reduces its undrained shear strength. Also, it is found that by increasing the clay content up to 10%, the undrained shear strength of reinforced and unreinforced clayey sand are decreased and brittleness index increased.

With further rises in clay content, the undrained shear strength increases and brittleness behavior decreases.

Keywords Liquefaction · Geotextile · Clayey sand · Steady state strength · Undrained triaxial test

1 Introduction

In recent years, undrained loading on the loose formation of soil deposits has led to considerable deformation on the soil surface and triggering damages on adjacent structure. Liquefaction is a destructive phenomenon in geotechnical engineering problems wherein the shear strength of the saturated soil drops sharply, and the pore water pressure reaches high values under undrained loading condition. Previously, It had been thought that only clean sands and sands containing small amount of non-plastic fines are subjected to liquefaction, thus so many researches has devoted to the undrained behavior of these soils (Castro and Poulos 1977; Kramer and Seed 1988; Lindenberg and Koning 1981; Lü et al. 2018; Sadrekarimi 2013; Seed and Lee 1966). The occurrence of liquefaction in the cases that the soil mass contained plastic fines, attracted so many attention on liquefaction of sand containing plastic fines (Abedi and Yasrobi 2010; Chang 1987; Kishida 1969; Miura et al. 1995; Wei and Yang 2014; Yang and Wei 2012). In 2002, one of the astounding cases observed in

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Changureh, Avaj-Iran earthquake that was a magnitude of 6.3 Richter. The surface of the soil was substantially clay; nonetheless, clear signs of sand boiling, soil softening, and triggering deformation were observed. The laboratory investigation indicated the soil contained a clay fraction of 44% (Ghahramani et al. 2006). Static liquefaction occurs under monotonic undrained loading, which is typically associated with loose saturated sand and sand-fines mixtures, defined as a large reduction in mean effective stress induced by high pore water generation (Poulos 1981). The decreasing trend of deviatoric stress continues until the soil reaches the lower shear strength at steady state condition. The steady state is defined as a condition when soil deforms under a constant void ratio, effective stress and shear stress (Fig. 1).

Numerous studies have been performed to evaluate static liquefaction potential and undrained behavior of clayey sand soils. The effect of clay particles on the undrained shear strength of sandy soils depends on several factors, such as plasticity of fines, clay content, clay mineral, and sand gradation. Several types of research have been devoted to examining these parameters (Bobei et al. 2009; Georgiannou et al. 1990; Naeemifar and Yasrobi 2011; Ovando-Shelley and Perez 1997; Pakbaz and Moqaddam 2012; Papadopoulou and Tika 2016; Pitman et al. 1994; Rahman and Lo 2014; Wood and Kumar 2000), although there is still some ambiguity in this field of study.

One of the most important factors which affects undrained behavior of sand-clay mixtures is clay

content. Previous researchers reported that an increase in the clay content up to range of 7 to 15%, causes reduction in shear strength and beyond these values, the shear strength climbs (Abedi and Yasrobi 2010; Bouferra and Shahrour 2004; Georgiannou et al. 1990; Gratchev et al. 2006, 2007). Georgiannou et al. (1990) by carrying out undrained triaxial shear tests indicated that the undrained brittleness rises as the clay content increases from 4.6 to 10%. Bouferra and Shahrour (2004) reported that the increase in the clay content up to a value of 15%, leads to a growth in the liquefaction potential of sand-clay mixture. It was stated that the observed behavior is due to the active role of clay content in reducing the dilatants tendency of sand particles. Gratchev et al. (2006, 2007) by performing a series of the scanning electron microscope (SEM) and cyclic triaxial tests on bentonite–sand mixtures, concluded that a small amount of bentonite in the soil mass, F.C. = 7%, causes to form an open micro fabric with low liquefaction resistance. It was observed that a bridge-like structure formed at the contact point of sand grains. An increase in clay content (F.C. = 15%) will cause to develop a more compact micro fabric substantially with limited pore space and higher resistance to liquefaction. Overall, it was concluded that the bentonite–sand mixture with low fines content is more vulnerable to liquefaction than clean sand.

Many efforts have been devoted to restricting the softening behavior of loose sand. Soil reinforcement is a relatively modern technique employed tensile elements to improving the shear strength of soil. Nowadays, geotextile due to features like high tensile

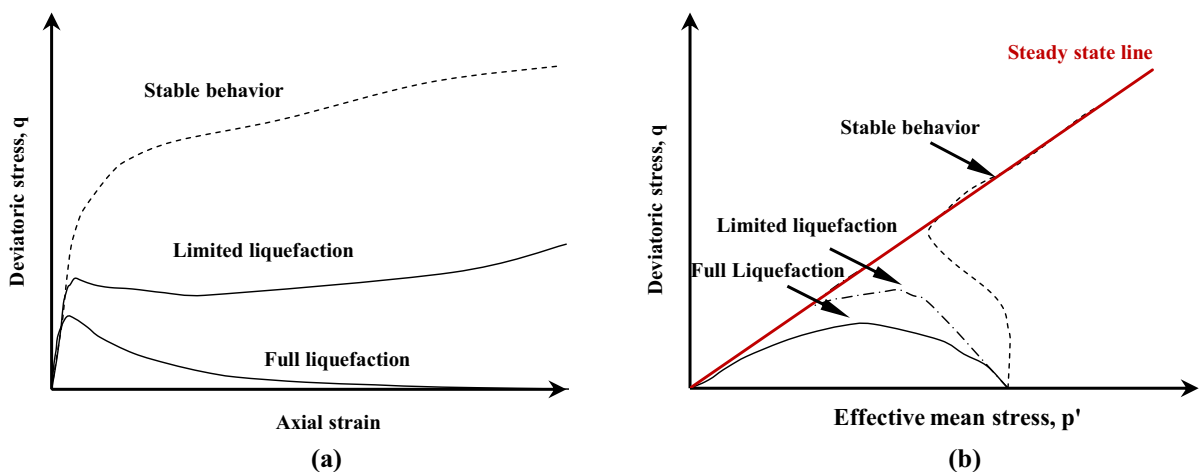


Fig. 1 Definition of static liquefaction

strength, high durability, low cost, and less environmental degradation, are used widely as reinforcement material (Aydilek et al. 2005; Ghosh and Bera 2010; Imjai et al. 2019; Lakirouhani and Bahrehdar 2018; Moghaddas Tafreshi et al. 2016; Perkins and Edens 2003; Yang et al. 2019; Zimmie et al. 2005; Zheng and Fox 2017; Zornberg et al. 2017). Some preliminary works were carried out by performing triaxial, plain strain and direct shear test to investigate the stress–strain and shear strength characterization of geotextile reinforced sand (Abdelkader et al. 2018; Afzali-Nejad et al. 2017; Ashmawy 1998; Athanasopoulos 1993; Broms 1977; Gray and Al-Refeai 1986; Goodhue et al. 2001; Haeri et al. 2000; Krishnaswamy and Thomas Isaac 1995; Lopes and Silvano 2010; Makiuchi and Miyamori 1988). The most obvious observation was the increase in shear strength. Furthermore, in some research, triaxial shear test results indicated the reinforced soil exhibited the lower post-peak reduction compared to the unreinforced soil (Benessalah et al. 2016; Broms 1977; Gray and Al-Refeai 1986; Goodarzi and Shahnazari 2019; Haeri et al. 2000; Markou 2016).

Krishnaswamy and Thomas Isaac (1995) performed a series of cyclic triaxial test on sands with different densities reinforced with nonwoven geotextile and reported that reinforcement causes an increasing in liquefaction resistance. It was also noted that the geotextile effect was more pronounced at lower relative densities. Moghaddas Tafreshi and Asakereh (2007) reported the influence of nonwoven geotextile layers was different at various axial strain levels, and geotextile reinforcement showed better performance at higher axial strains. Habibi et al. (2014) outlined that woven geotextile inclusion seems to be more effective in the sand rather than clay, in other words, sand particles showed better interlocking with geotextile layers. To evaluate the effect of geotextile arrangement in a triaxial specimen, by performing triaxial cyclic tests, it was concluded that as the nearest geotextile layer to the specimen surface, it has more impact on the improvement of axial modulus in cyclic loading conditions (Moayed and Alibolandi 2018; Naeini and Gholampoor 2014).

According to presented literature, most of the previous studied has focused on the effect of geotextile reinforcement on shear strength characteristic of sandy soils. Hence, this paper investigates the performance of geotextile inclusion in enhancing the

undrained behavior of loose clayey sand. For this purpose, a series of consolidated undrained triaxial tests were performed on unreinforced and geotextile reinforced sand-clay mixtures with various clay contents. The triaxial tests were performed on three geotextile arrangements. The discussions are mostly about strain-softening and steady state strength behavior. Also, the analysis of undrained shear strength and how it is affected by geotextile reinforcement in different clay contents will also be discussed.

2 Experimental Work

2.1 Material Properties

The clayey sand tested in this research comprises of clean Firoozkooh 161 sand and kaolin. The 161 sand is clean fine silica sand which is categorized as poorly graded with uniform gradation sand (SP) according to the United Soil Classification System (USCS). The kaolin fraction is low plasticity clay with 20% silt and a plasticity index of 21. The physical properties of 161 sand and kaolin are summarized in Table 1, and the particle size distribution curves are shown in Fig. 2. It can be seen by increasing the fines content in the soil, the mean particle size (D_{50}) decreased and particle size distribution ranges expanded.

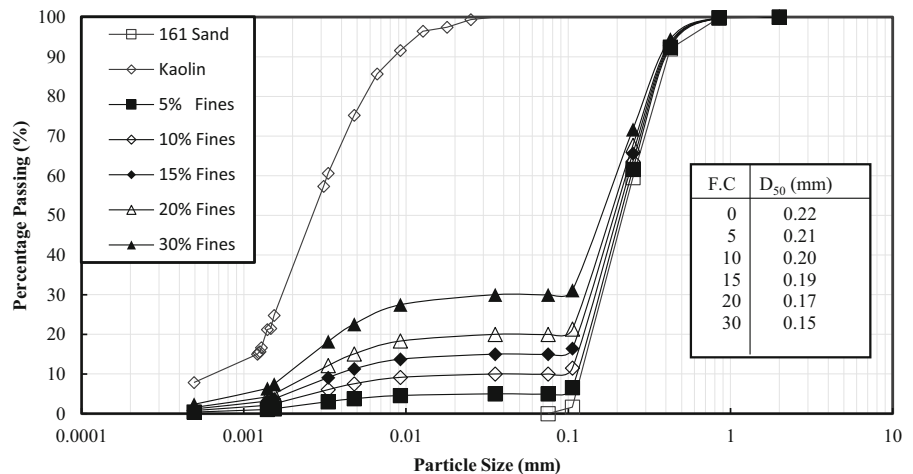
Based on the research background, it was concluded that the utilization of geotextiles with a minimum weight of 120 g/m² and tensile strength of 8 kN can be effective in enhancing soil strength (Goodarzi and Shahnazari 2019; Gray and Al-Refeai 1986; Haeri et al. 2000; Markou 2016; Moghaddas Tafreshi and Asakereh 2007). Nonwoven needle punched geotextile supplied by an Iranian geosynthetics manufacturer was used in the experimental programs (Fig. 3). The physical and mechanical properties of the geotextile obtained by the company are presented in Table 2.

2.2 Specimens Preparation

All test specimens were prepared using moist tamping technique at the water content of 6%. The moist soil was compacted in eight equal layers with a suitable tamper. All tests were conducted on specimens with 70 mm diameter and 140 mm height. All mixtures were prepared in a dry unit weight of 14.35 kN/

Table 1 Physical properties of sand and kaolin

Soil	Specific gravity, G_s	Min void ratio, $e_{\min}$	Max void ratio, $e_{\max}$	Uniformity coefficient, C_u	Gradation coefficient, C_z	Liquid limit, LL (%)	Plastic limit, PL (%)	Plasticity index, PI (%)
Sand	2.658	0.59	0.91	1.9	0.89	–	–	–
Kaolin	2.589	–	–	–	–	48	27	21

Fig. 2 Grain size distribution curve of sand, kaolin and sand-kaolin mixtures**Fig. 3** Used geotextile

m^3 . It should be mentioned that according to the loose density of tested soil, no compaction test were performed.

In order to investigate the effect of clay content on the behavior of sand, the specimens were made with 0, 5, 10, 15, 20, and 30% kaolin content. For evaluation of the effect of geotextile inclusion, specimens were reinforced with three various geotextile arrangements

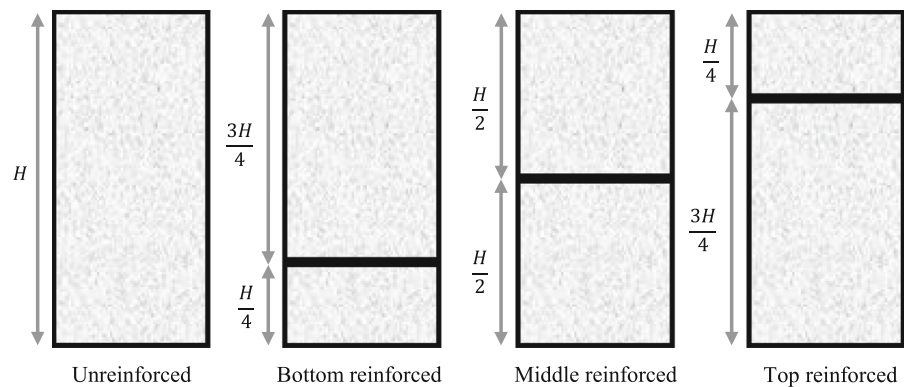
Table 2 Mechanical properties of geotextile

Properties	Value
Type	Nonwoven-needle punch
Structure	Continuous filament
Mass per unit area (g/m^2)	250
Nominal thickness (mm)	3
Maximum tensile strength (kN/m)	16
Puncture resistance (kN)	3
Permittivity (s^{-1})	1.45

of bottom, middle, and top reinforced. The arrangements of geotextile are presented in Fig. 4.

2.3 Test Procedure

In order to investigate the undrained behavior of reinforced clayey sand, a total of 72 consolidated undrained triaxial tests were performed according to ASTM D4767-11 (2011) on unreinforced and reinforced specimens with different arrangements of geotextile layer.

Fig. 4 Geotextile arrangements

Saturation of specimens was accomplished by percolating of CO_2 and then de-aired water from the bottom to top of the specimen. Specimens were considered as fully saturated when the B parameter was equal or greater than 0.95. The specimens were consolidated under three different effective confining pressures of 100, 150, and 200 kPa. After the consolidation phase, the specimens were subjected to compression loading using a constant rate of 0.5 mm/min at undrained condition. The rate of axial strain was determined according to ASTM D4767, which 0.5 mm/min is corresponded to the mixture containing 30% clay content (the lowest strain rate among all mixtures). This rate was used to provide approximate equalization of pore pressure throughout the specimen.

3 Results and Discussion

A series of undrained triaxial tests were performed on isotropically consolidated unreinforced and reinforced clayey sands. Experiments were carried out on specimens with different geotextile arrangements and different clay contents. The effect of plastic fine sand reinforcement inclusion on the undrained behavior of clayey sands will be discussed, as followed. Table 3 presents the summary of triaxial test results.

3.1 Effect of Clay Content on Undrained Behavior

Figure 5 presents the variation of deviatoric stress and pore water pressure versus axial strain and stress path for unreinforced specimens with various fines contents under 150 kPa confining pressure. The general trend in stress–strain and pore water pressure of mixtures

were approximately similar for various confining pressures preferred to present the graph of confining pressure 150 kPa as a representative.

It can be seen that the clean sand exhibited a dilative behavior (Fig. 5a); as the contraction of the sand grains occurs through the small void space, the pore water pressure increased at low strains ($\epsilon < 1.5\%$), and afterward because of the negative pore water pressure generation, it declined gradually. The negative pore pressure generated through sand specimens (Fig. 5b) is caused by the sand grains movement on each other. In such specimens, the stress path exhibited a hardening tendency after the initial contractive behavior (Fig. 5c), which the effective stress increased due to the negative pore water pressure generation (Fig. 5b).

The different behavior of clayey sand and clean sand specimens can be clearly seen in Fig. 5, where the clayey soil displayed a contractive behavior, so that the stress–strain curves reached a peak value, and then dropped to the steady state strength. The contractive behavior is attributed to the accumulation of clay particles at the contact surface of the sand grains (Gratchev et al. 2007) and the high compressibility of clay particles (Mitchell 1976). As shown in Fig. 5b, positive pore water pressures developed during tests on clayey sand specimens, then reached a steady state value at large strains.

As illustrated in Fig. 5c, clayey sand specimens tolerated strain-softening behavior under the same confining pressures. The amount of clay fraction transformed static liquefaction potential of clayey sand specimens. By raising the fines content up to 10%, the minimum value of mean effective stress was decreased. By further additions in clay content, the

Table 3 Summary of triaxial test results

Test	Effective Confining pressure, σ'_3 (kPa)	Fines content, Fc (%)	Geotextile arrangement	Void ratio after consolidation, e_f	Steady State Undrained shear strength, q_{ss} (kPa)	Steady state mean effective stress, p'_{ss} (kPa)	Peak Undrained shear strength, q_{peak} (kPa)
S-U-1	100	0	Unreinforced	0.806	190	145	190
S-U-2	150	0	Unreinforced	0.801	240	180	240
S-U-3	200	0	Unreinforced	0.799	285	216	285
S-Rm-1	100	0	Middle	0.801	281	190	281
S-Rm-2	150	0	Middle	0.797	360	246	360
S-Rm-3	200	0	Middle	0.792	473	328	473
S-Rt-1	100	0	Top	0.803	336	245	336
S-Rt-2	150	0	Top	0.797	437	324	437
S-Rt-3	200	0	Top	0.793	510	376	510
S-Rb-1	100	0	Bottom	0.802	349	261	349
S-Rb-2	150	0	Bottom	0.797	466	339	466
S-Rb-3	200	0	Bottom	0.792	625	454	625
SF5-U-1	100	5	Unreinforced	0.801	17	15	66
SF5-U-2	150	5	Unreinforced	0.794	37	29	93
SF5-U-3	200	5	Unreinforced	0.788	57	49	118
SF5-Rm-1	100	5	Middle	0.793	43	31	73
SF5-Rm-2	150	5	Middle	0.784	64	48	97
SF5-Rm-3	200	5	Middle	0.778	105	72	135
SF5-Rt-1	100	5	Top	0.795	31	24	70
SF5-Rt-2	150	5	Top	0.787	57	40	105
SF5-Rt-3	200	5	Top	0.776	100	74	138
SF5-Rb-1	100	5	Bottom	0.794	48	38	73
SF5-Rb-2	150	5	Bottom	0.783	64	55	110
SF5-Rb-3	200	5	Bottom	0.780	116	80	145
SF10-U-1	100	10	Unreinforced	0.774	1	1	32
SF10-U-2	150	10	Unreinforced	0.761	2	3	62
SF10-U-3	200	10	Unreinforced	0.751	7	9	87
SF10-Rm-1	100	10	Middle	0.772	2	2	39
SF10-Rm-2	150	10	Middle	0.758	7	4	65
SF10-Rm-3	200	10	Middle	0.746	25	17	94
SF10-Rt-1	100	10	Top	0.774	4	5	36
SF10-Rt-2	150	10	Top	0.757	9	8	67
SF10-Rt-3	200	10	Top	0.751	23	23	93
SF10-Rb-1	100	10	Bottom	0.771	7	5	41
SF10-Rb-2	150	10	Bottom	0.766	20	19	77
SF10-Rb-3	200	10	Bottom	0.754	32	21	103
SF15-U-1	100	15	Unreinforced	0.706	3	4	33
SF15-U-2	150	15	Unreinforced	0.686	14	12	62
SF15-U-3	200	15	Unreinforced	0.676	19	17	82
SF15-Rm-1	100	15	Middle	0.715	7	8	41
SF15-Rm-2	150	15	Middle	0.694	29	20	70
SF15-Rm-3	200	15	Middle	0.686	45	34	91

Table 3 continued

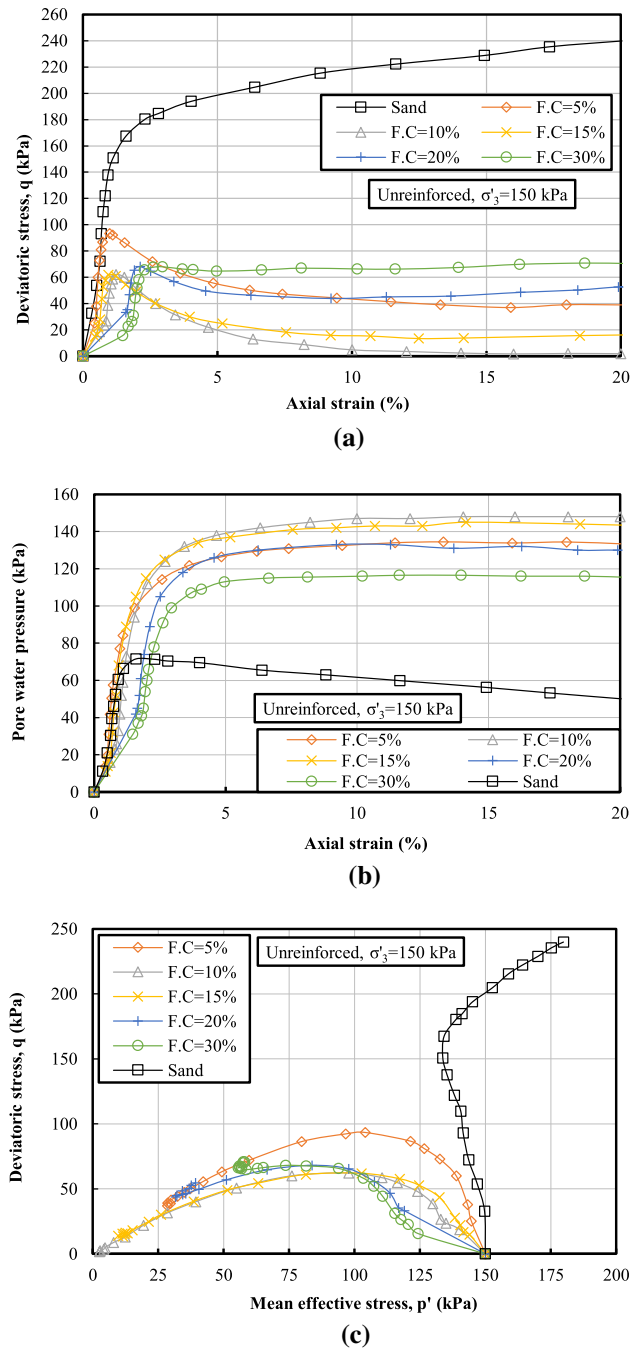
Test	Effective Confining pressure, σ'_3 (kPa)	Fines content, Fc (%)	Geotextile arrangement	Void ratio after consolidation, e_f	Steady State Undrained shear strength, q_{ss} (kPa)	Steady state mean effective stress, p'_{ss} (kPa)	Peak Undrained shear strength, q_{peak} (kPa)
SF15-Rt-1	100	15	Top	0.716	15	9	41
SF15-Rt-2	150	15	Top	0.710	25	19	65
SF15-Rt-3	200	15	Top	0.696	43	34	88
SF15-Rb-1	100	15	Bottom	0.711	16	8	44
SF15-Rb-2	150	15	Bottom	0.705	31	17	76
SF15-Rb-3	200	15	Bottom	0.699	53	33	101
SF20-U-1	100	20	Unreinforced	0.640	22	18	36
SF20-U-2	150	20	Unreinforced	0.624	44	32	68
SF20-U-3	200	20	Unreinforced	0.612	62	46	89
SF20-Rm-1	100	20	Middle	0.626	32	28	41
SF20-Rm-2	150	20	Middle	0.618	49	33	73
SF20-Rm-3	200	20	Middle	0.609	67	44	100
SF20-Rt-1	100	20	Top	0.643	27	18	43
SF20-Rt-2	150	20	Top	0.614	51	38	69
SF20-Rt-3	200	20	Top	0.602	78	54	100
SF20-Rb-1	100	20	Bottom	0.624	34	23	44
SF20-Rb-2	150	20	Bottom	0.621	58	40	79
SF20-Rb-3	200	20	Bottom	0.613	77	51	106
SF30-U-1	100	30	Unreinforced	0.598	41	28	49
SF30-U-2	150	30	Unreinforced	0.592	66	56	68
SF30-U-3	200	30	Unreinforced	0.574	85	63	97
SF30-Rm-1	100	30	Middle	0.605	46	40	50
SF30-Rm-2	150	30	Middle	0.584	77	59	77
SF30-Rm-3	200	30	Middle	0.581	92	73	103
SF30-Rt-1	100	30	Top	0.614	48	43	50
SF30-Rt-2	150	30	Top	0.588	68	55	74
SF30-Rt-3	200	30	Top	0.568	100	81	103
SF30-Rb-1	100	30	Bottom	0.611	51	40	52
SF30-Rb-2	150	30	Bottom	0.596	74	55	78
SF30-Rb-3	200	30	Bottom	0.581	100	76	108

mean effective stress increased and soil underwent less instability.

Figure 6 shows the amount of clay fraction affected the undrained shear strength of clayey sand specimens. For unreinforced samples under all confining pressures, as the fines content increased up to a threshold value (about 10%), the steady state strength decreased, by further increase of fines content, and the steady state strength increased. Up to 10% clay contents, the mixture has an open micro fabric with very low shear

resistance. An increase in clay content (more than 10%) will cause to form a more compact micro fabric with limited pore space which subsequently sustains higher shear stress (Gratchev et al. 2006, 2007). Comparison of pore water pressure generation proves an unstable structure of the specimen with 10% fines content (Fig. 5b); meanwhile, the excess pore water reached to a steady state with a value of 148 kPa under confining pressure of 150 kPa. This behavior indicates the unstable formation of soil particles that is

Fig. 5 Effect of fines content on the undrained behavior of unreinforced sand–clay mixtures under confining pressure of 150 kPa: **a** deviatoric stress versus axial strain; **b** pore water pressure versus axial strain; **c** effective stress paths

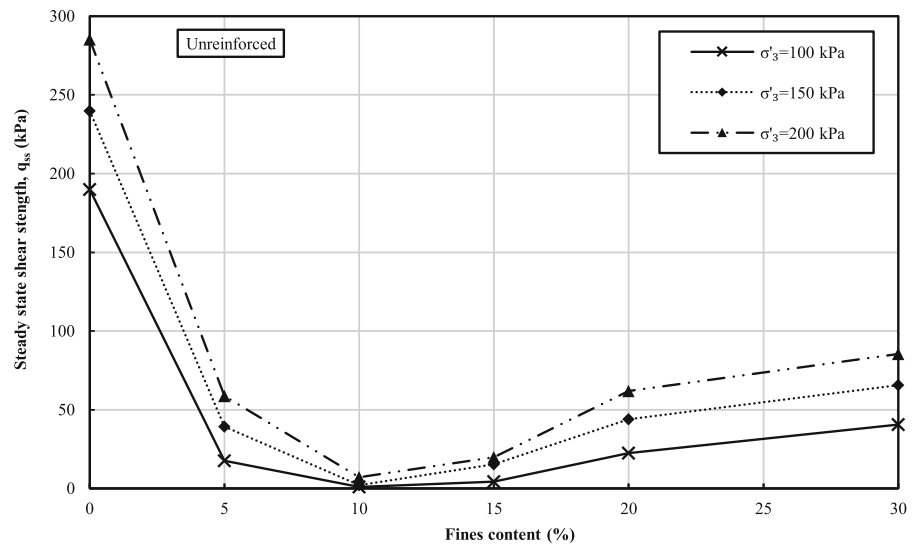


remarkably close to the full liquefaction behavior and as can be seen from Figs. 5c and 6, for mixture with 10% fines content, mean effective stress is close to zero. In such behavior, as the soil structure is destroyed, the increase of confining pressure leads to increase the pore pressure instantly.

3.2 Effect of Geotextile on Undrained Behavior

When the geotextile layer is used as an inclusion, the undrained behavior of reinforced specimens appears far better than unreinforced specimen in enhancing shear strength and reducing the post-peak loss of strength. Figure 7 presents the undrained response of

Fig. 6 Variation of the steady state strength with increase in the fines content



bottom reinforced specimens with different fines contents under confining pressure of 150 kPa. By comparing Figs. 5 and 7, it can be stated that, for various sand-clay mixtures under the same confining pressure, the overall behavior of unreinforced and reinforced specimens were similar, but reinforcement caused an improvement in steady state and peak shear strength of the specimens. In other words, reinforced specimens similar to unreinforced specimens exhibited dilative behavior in clean sand and contractive behavior in clayey sands, but the differences were in the strength values that were higher for reinforced soils. Moreover, using geotextile led to decrease post-peak instability of clayey sand specimens and increase the minimum mean effective stress.

Reinforcing the soil specimens with geotextile layer caused to form an interaction between soil and geotextile. The formed interaction resulted in enhancing the shear strength of the soil by providing excess confinement in soil caused by geotextile. Figure 8 presents the stress–strain curve, pore water pressure and stress paths of unreinforced and reinforced clean sand specimens with various arrangements of geotextile under 200 kPa confining pressure. It can be seen that employment of geotextile layer in sandy specimens causes an improvement in undrained shear behavior. For example, the deviatoric stress of unreinforced sand specimen at 20% axial strain is 288 kPa, while for specimens with bottom, top and middle geotextile arrangements are 626, 510 and 473 kPa, respectively. As can be seen in Fig. 8b, reinforcement

led to decrease the pore water pressure in soil specimen. The reduction of pore water pressure in reinforced soils indicates internal drainage occurred in reinforced specimens. Furthermore, geotextile reinforcement led to improve hardening behavior of clean sand specimen, as stress path of unreinforced sand extended to higher mean effective stress by reinforcing with geotextile (Fig. 8c).

The strengthening of geotextile reinforcement increases as the shear test progresses. In other words, the mobilization of geotextile interface formed more effective at high strains. As an example, the bottom reinforcement strengthened the deviatoric stress by 64% at the strain of 5%, whereas it increased by 117% at the strain of 20%. This enhancement is due to the proper interaction between the geotextile and soil particles at high strains; as the test progressed, the soil particle moved toward each other and formed interlocking with the geotextile layer. These findings are in good agreement with the results of Moghaddas Tafreshi and Asakereh (2007).

Similar to sand specimens, the employment of the geotextile layer causes to increase the undrained shear strength in clayey sand specimens. In order to display the effect of geotextile on the undrained behavior of clayey sand, only the stress–strain, pore water pressure and stress path of clayey sand with F.C. = 5% under 200 kPa of confining pressure is shown as a representative in Fig. 9 and the comparison of clay-sand mixtures behavior are discussed in Sects. 3.2.1 and 3.3. The stress–strain curves for unreinforced and

Fig. 7 Effect of fines content on the undrained behavior of bottom reinforced sand–clay mixtures under confining pressure of 150 kPa: **a** deviatoric stress versus axial strain; **b** pore water pressure versus axial strain; **c** effective stress paths

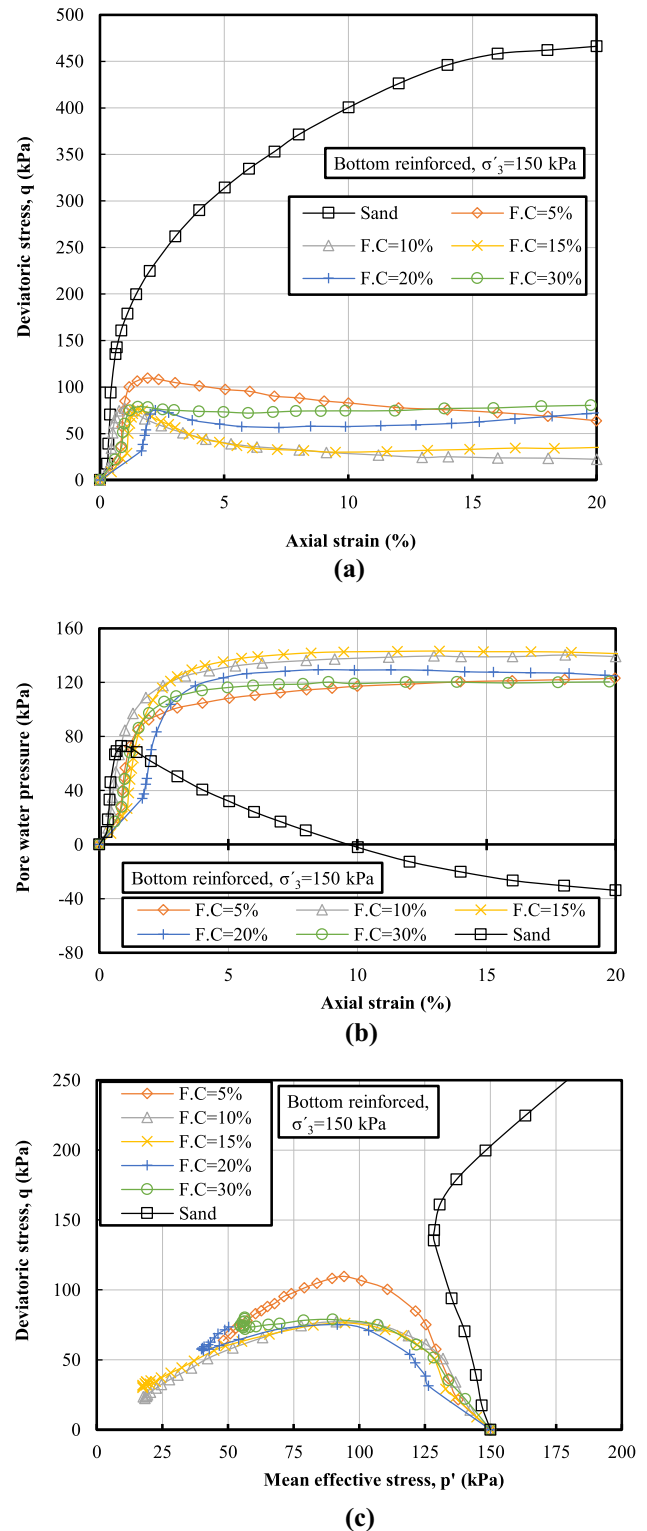
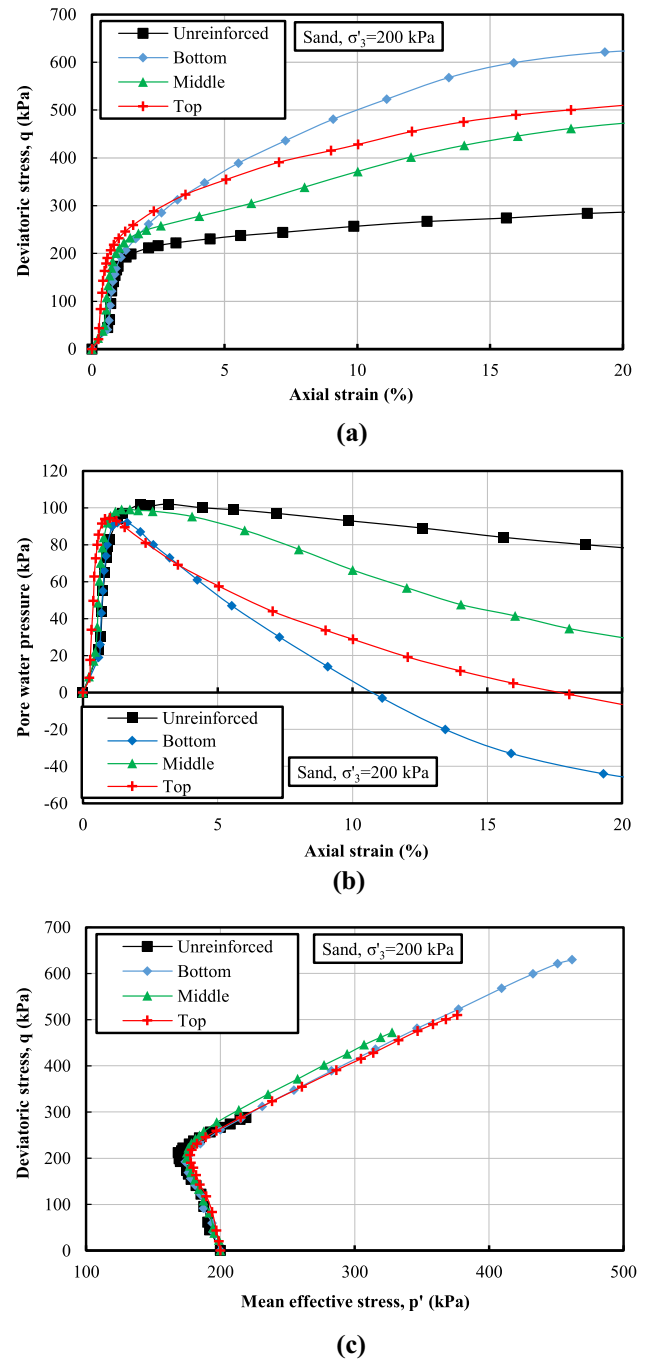


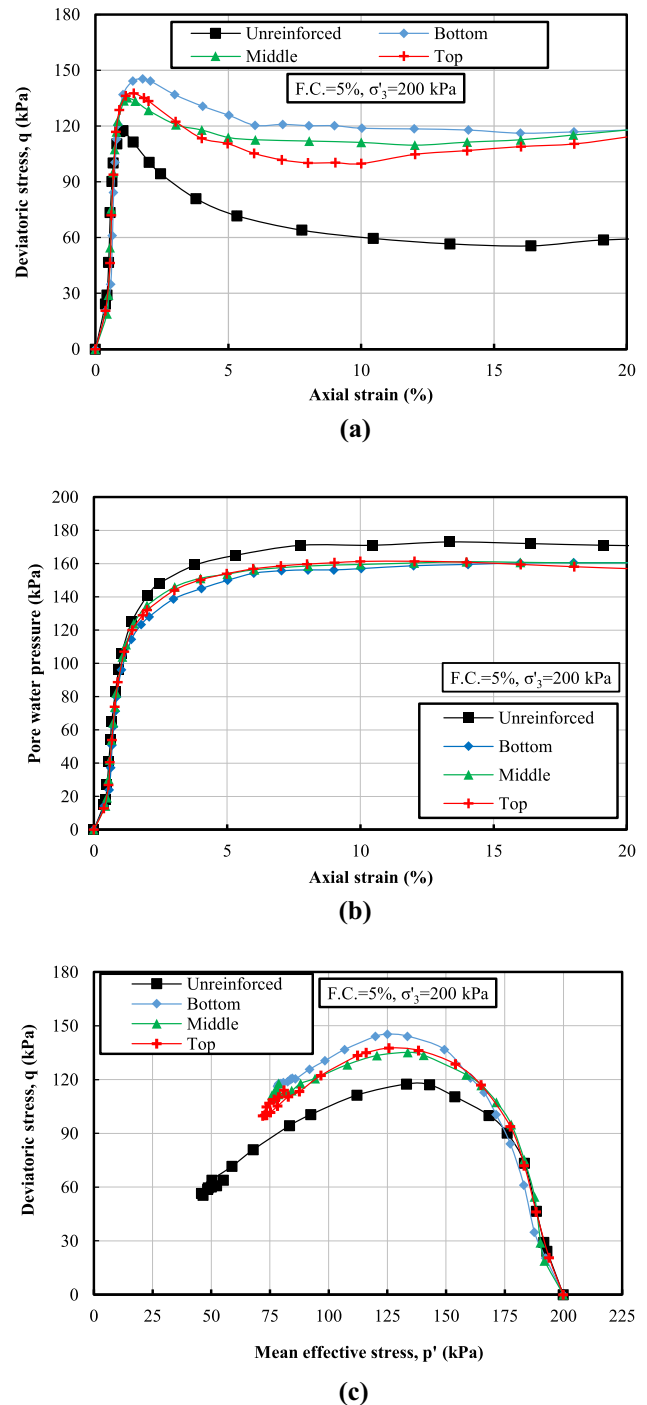
Fig. 8 Plots of **a** stress–strain curves; **b** pore water pressure versus axial strain; **c** effective stress paths for unreinforced and reinforced sand under 200 kPa confining pressure



reinforced clayey sand are so close to each other at low strain ($\varepsilon < 1\%$), and the peak shear strength of the reinforced specimens are larger than the unreinforced one (Fig. 9a). The effect of geotextile on the shear strength of clayey sands is observed at higher strains. It is worth noting that geotextile reinforcement caused to

decrease in the post-peak instability tendency of clayey sand specimens. In reinforced specimens less positive pore water pressure was generated that led to higher mean effective stress at steady state (Fig. 9b). As shown in Fig. 9c, the marked reduction in post-peak strength of sand with 5% fines content decreases

Fig. 9 Plots of **a** stress–strain curves; **b** pore water pressure versus axial strain; **c** effective stress paths for unreinforced and reinforced sand containing 5% fines content under 200 kPa confining pressure



by reinforcing with geotextile layer under the same confining pressure. For example, in the unreinforced specimen with F.C. = 5%, the peak deviatoric strength reached 117 kPa after 1% axial strain, then with a strain-softening behavior dropped to a steady state

strength of 57 kPa. While reinforcing the specimen led to decrease in the instability occurring after peak strength, as the deviatoric stress of bottom reinforced specimen dropped from the peak strength of 145 kPa to the steady state strength of 116 kPa. This

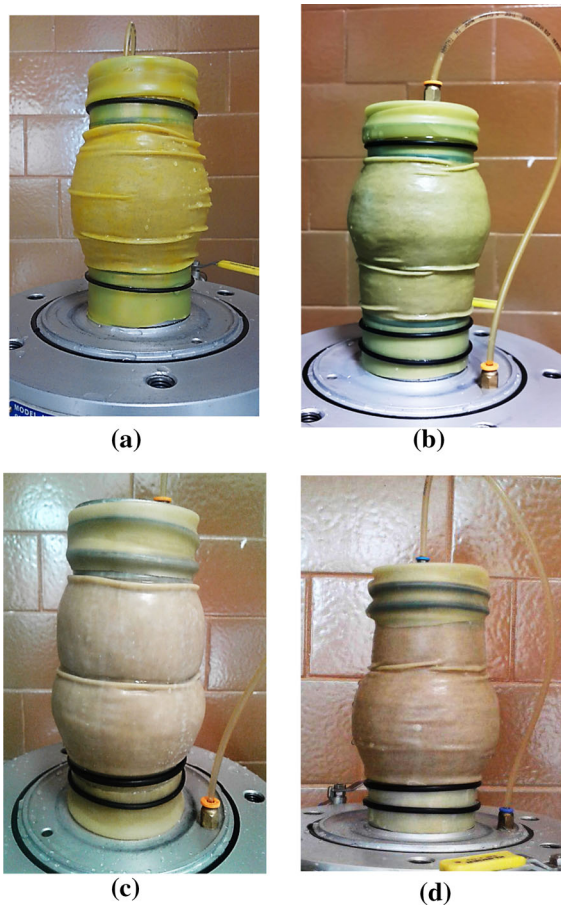


Fig. 10 Images of failure modes of: **a** unreinforced; **b** bottom reinforced; **c** middle reinforced; **d** top reinforced sand

performance means that geotextile reinforcement leads to limit the static liquefaction potential.

The images of deformed sand specimens at the end of the test are presented in Fig. 10 for unreinforced and reinforced clean sand. For unreinforced sand specimens, the specimen deformed laterally in a barrel shape (Fig. 10a). While geotextile inclusion changed the lateral deformation so that by placing the geotextile at $\frac{1}{4}$ bottom of the specimen, the upper part deformed in a barrel shape and the lower part had almost no deformation (Fig. 10b); by placing the geotextile at the middle of the specimen, the specimen deformed as two barrel shape from the geotextile level (Fig. 10c); and by placing the geotextile at $\frac{1}{4}$ top of the specimen, the lower part deformed in a barrel shape and the upper part almost had no lateral deformation (Fig. 10d). It can be deduced from the specimens' deformation that the geotextile inclusion

led to restrict the lateral expansion at the level where the geotextile located, which confirmed the effective mobilization of geotextile and soil interfaces.

3.2.1 Effect of Geotextile Reinforcement on Steady State Strength

Figure 11 shows the steady state strength of reinforced and unreinforced sand specimens under various confining pressures. For the sand specimens, the stress corresponded to the axial strain of 20% was considered as steady state shear strength. It can be stated that the steady state shear strength of the reinforced sand specimens are higher than unreinforced sand specimen under all confining pressures. It is also worth noting that the undrained shear strength was greater under higher confining pressures.

To carry out a comparative investigation on how geotextile affects the shear strength of various sand-clay mixtures, a parameter of *steady state strength ratio*, R_{ss} , is defined as Eq. (1):

$$R_{ss} = \frac{q_{ss, \text{reinforced}}}{q_{ss, \text{unreinforced}}} \quad (1)$$

where $q_{ss, \text{reinforced}}$ and $q_{ss, \text{unreinforced}}$ are the steady state deviatoric stress of reinforced and unreinforced specimens, respectively. The higher the R_{ss} value for a specimen, the better the geotextile performance in such a specimen.

Figure 12 presents steady state shear strength ratio, R_{ss} , versus fines content under 200 kPa confining pressures. It is found that the performance of the geotextile layer is more effective at the mixture with low fines contents. This behavior is because of the

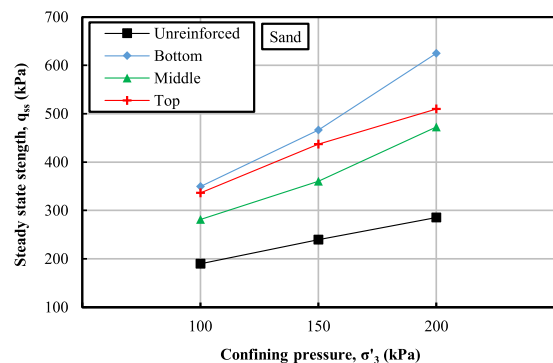
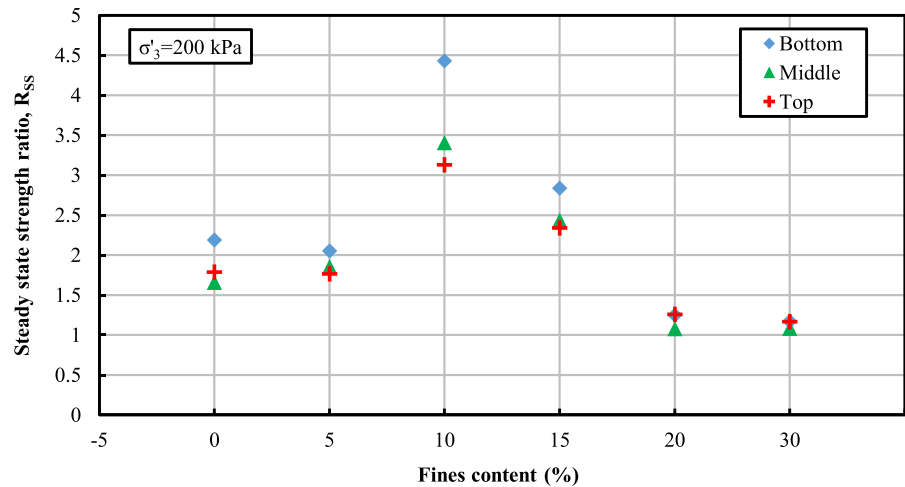


Fig. 11 Effect of geotextile reinforcement on steady state strength of sand

Fig. 12 Variation of steady state ratio for various mixtures under 200 kPa confining pressure



proper interaction between sand particles and geotextile layer; the more sand grains in the specimen, the better interaction between soil grains and geotextile. The values of R_{ss} for the specimens with 30% fines content is close to 1, which indicates a poor interaction between geotextile and soil containing high values of fines content. These observations lend support to the previous finding in the literature by Habibi et al. (2014).

Noticeable improvement of shear strength is evident for specimens with the fines content of 10 and 15%; in these mixtures, the values of R_{ss} were higher than other specimens, while these two mixtures have the lowest values of steady state shear strength among all mixtures. The geotextile reinforcement at bottom led to increase the steady state shear strength of the sand specimen under confining pressure of 200 kPa from 285 to 625 kPa, and the value of R_{ss} at this case is equal to 2.19, while the reinforcement in the same arrangement and confining pressure for clayey sand specimen with F.C. = 10%, increases the steady state shear strength from 7 to 32 kPa and the value of R_{ss} is equal to 4.5.

The results revealed that among all mixtures, the best efficiency of geotextile arrangement in improving the steady state shear strength is observed in bottom reinforced specimens (Fig. 12). In this study, the results did not show enough consistent trend between shear strength of middle and top reinforced specimens among all sand-clay mixtures.

3.3 Brittleness Index

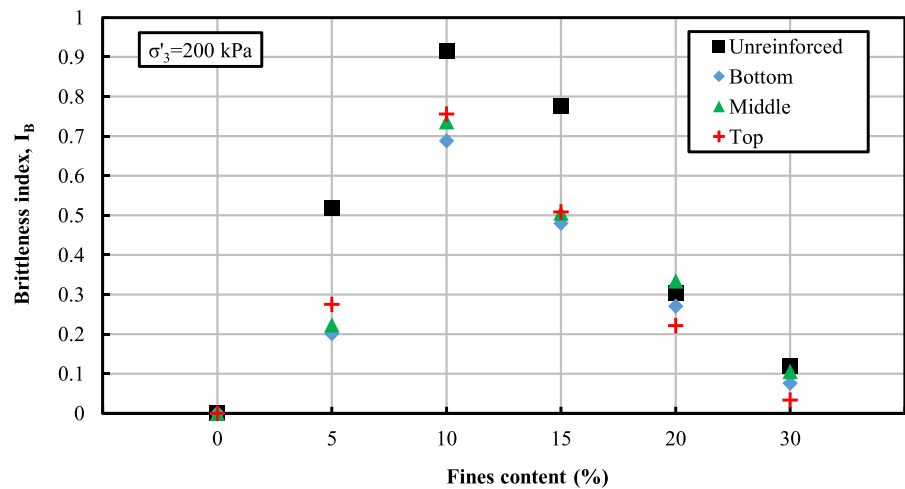
As observed in stress–strain curves (Figs. 5 and 7), the mixtures containing clay content exhibited a strain-softening behavior; after the peak shear stress, an instability occurred in the specimen and undrained shear strength dropped to a steady state. The reduction of strength for a strain-softening material can be expressed in term of the *Brittleness index*, I_B (Bishop 1967), and can be calculated by Eq. (2):

$$I_B = \frac{q_p - q_{ss}}{q_p} \quad (2)$$

where q_p is the peak undrained shear strength, and q_{ss} is the steady state undrained shear strength. The brittleness index could be a good criterion for evaluating the static liquefaction potential for contractive soils (Keramatikerman et al. 2018; Sabbar et al. 2017; Sadrekarimi 2014, 2016). The values of the brittleness index are in a range of 0 to 1. As the soil material has greater strength reduction, the brittleness index value is closer to 1. Reaching the strain-softening state leads to large deformations for materials with high values of the brittleness index. Undrained behavior of soil with $I_B = 0$ defined as non-flow behavior.

As illustrated in Fig. 13, the presence of clay fines attenuates the soil hardening behavior, and consequently increases its undrained brittleness. This result is clearly observed for specimens with low fines content, and become more pronounced when the fines content increases to 10%. Beyond this value, the clayey sand exhibited other trends; the increase in fines content causes a ductile behavior on the soil

Fig. 13 Effect of geotextile reinforcement on the brittleness index



which tolerated lower unstable behavior after peak strength; therefore, a decrease in the brittleness index is observed. According to Fig. 13, the values of brittleness index for unreinforced mixtures of 30% fines content under 200 kPa confining pressure is 0.12, which implies that the liquefaction potential of high fines content mixture is insignificant. In specimens of 30% fines content, there is a strain hardening behavior following the steady state strain (Fig. 5). This finding is consistent with the past study of Pitman et al. (1994), who reported that the undrained brittleness decreased as the plastic fines increased, and only strain hardening was observed towards the steady state in 40% fines content soil, in other words, there was no brittle behavior. Generally, based on the brittleness index values and stress–strain curves, it can be concluded that the sand containing 10% fines content is more susceptible to static liquefaction among all mixtures, this mixture exhibited large deformation, post-peak reduction and excess pore pressure under undrained monotonic loading and the behavior was so close to full liquefaction.

It was observed that soil reinforcement led to lessen strain-softening and decrease the brittleness behavior of soil specimens. The reinforced specimens had lower values of the brittleness index rather than unreinforced specimens (Fig. 13). For example, the reinforcement at the bottom layer caused a decrease in the brittleness index of the specimen with F.C. = 10% from 0.92 to 0.69. Geotextile reinforcement resulted in reducing the post-peak loss of strength and more ductility in specimens. It is also worth noting that the effectiveness of the geotextile layer on enhancement of ductile

behavior is insignificant for mixture with higher fines content; for mixture with F.C. = 30%, even though the brittleness index of unreinforced specimen is equal to 0.12, the brittleness index of bottom, middle and top reinforced specimens are equal to 0.08, 0.10 and 0.03, respectively.

As mentioned previously, at high strains, reinforcement enhances the undrained shear strength significantly, and at low strains, the undrained shear strength of reinforced and unreinforced specimens is so close to each other. For this reason, the values of the peak undrained shear strength of reinforced and unreinforced are close to each other, while due to the excess confinement induced by geotextile at high strains, the values of steady state undrained shear strength for reinforced specimens are greater than unreinforced specimens. Consequently, with respect to Eq. 2, it can be stated that the values of the brittleness index of the reinforced specimens are lower than unreinforced specimens. In other words, geotextile reinforcement by controlling the post-peak instability of loose soil, reduce the liquefaction potential of clayey sand. Other researchers stated that reinforcement reduces the post-peak shear loss of strength (Broms 1977; Gray and Al-Refeai 1986; Haeri et al. 2000).

3.4 Steady State Line

The steady state line of reinforced and unreinforced mixtures with various fines content in the p' – q plot are presented in Fig. 14. It can be seen that, in all mixtures, geotextile reinforcement led to an increase in the slope of the steady state line. Thus, it concluded

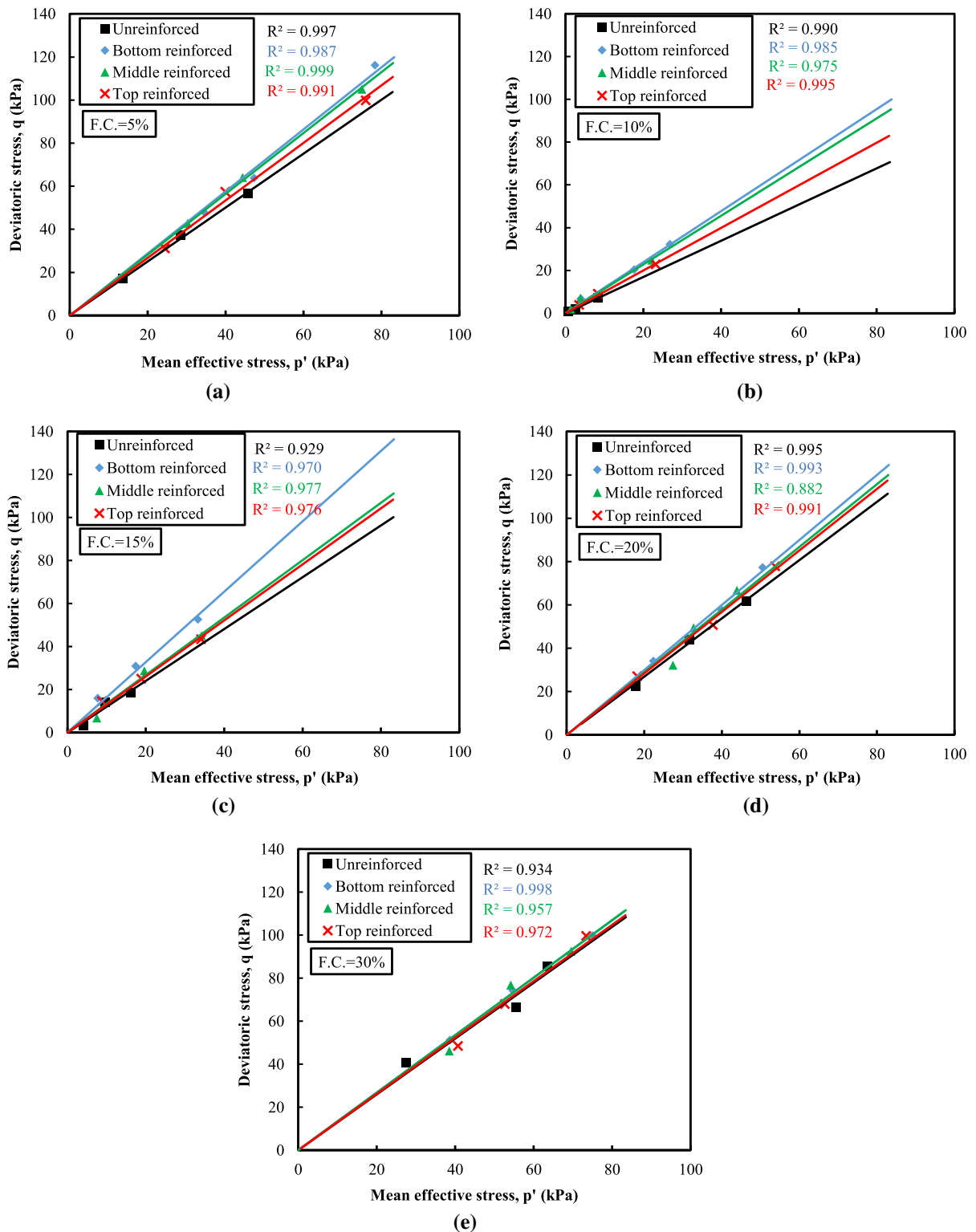


Fig. 14 Steady state line in p' - q plot

that geotextile inclusion improved the static liquefaction resistance of soil specimens. In other words, in an undrained condition, saturated reinforced soil compared to unreinforced one at a similar depth and void ratio, sustain higher deviatoric stress after the occurrence of static liquefaction. Figure 14 indicates the arrangement of geotextile could modify the slope of the steady state line. For all mixtures, the steady state line for bottom reinforced soil is steeper than other reinforcement arrangements in an undrained condition. The reason for this behavior can be seen in the development of proper interaction between soil particle and geotextile in instable condition during undrained loading.

As shown in Fig. 14, the effectiveness of reinforcement is variable for different mixtures. Up to a fines content between 10 and 15%, as the amount of clay fraction increased in soil specimen, the difference between the slope of steady state line for unreinforced and reinforced soil became more impressive, but with the further increase in fines content, the effect of reinforcement became negligible. In mixture with 10% and 15% fines content, given that the soil formation was an unstable structure, soil particles tended to develop large lateral deformation, therefore sand-geotextile interaction formed properly.

4 Conclusion

The undrained behavior of clayey sands and how geotextile inclusion effects on the shear strength of these mixtures were investigated by performing a series of consolidated undrained triaxial shear tests. The results show that:

- The clay particle changes the dilative behavior of unreinforced and reinforced sand specimen to contractive behavior. By increasing the amount of fines content in the soil to a value of 10%, the residual shear strength decrease, and a further increase in fines content increases the residual shear strength. The mixture with 10% fines content has the lowest residual strength and liquefaction resistance.
- The brittleness index increases by increasing the fines content up to 10%, but with further increase in fines content, the brittleness index decreases.
- Nonwoven geotextile reinforcement significantly enhances the undrained shear behavior of sand–clay mixtures.
- The geotextile efficiency decreases at mixtures with high fines content. It is due to the weak interlocking between clay particles and geotextile.
- Among all mixtures, geotextile reinforcement is more pronounced in mixture with 10% fines content. The reason for this performance is attributed to large movement of soil particles.
- Geotextile reinforcement caused a reduction in the brittleness index of sand mixed with varying amounts of clay under different confining pressures. This behavior is due to the proper interaction between soil grain and geotextile layer and providing excess confinement by geotextile at high strains.
- The slope of the steady state line increases with reinforcing the soil, which represents higher post-peak undrained shear strength of reinforced soils.
- Among various geotextile arrangements in all mixtures, bottom reinforcement found to be more effective in improving the undrained shear strength characteristics under all confining pressure, and there were not enough consistent trend between top and middle reinforced specimens among all sand–clay mixtures.

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